

EDUSUITE ENTERPRISE SOLUTIONS

EduSuite Pro ERP

Enterprise Educational Management Platform

PROJECT DOCUMENTATION

Project Title:	EduSuite Pro ERP System
Submitted To:	EduSuite Tech Solutions
Lead Developer:	Hanish (Employee ID: EMP-2026-809)
Development Team:	Keerthi, Vardhini, Preethi, Hemanth, Lokesh, Ramesh, Murali, Vishnu, Ashok, Satya
Department:	Enterprise Software Engineering & AI Solutions
Primary Tech Stack:	React 19, TypeScript, TanStack Start/Router, Vite, Recharts, Cloudflare Nitro
Submission Date:	06/08/2026
System Version:	v2.4.0 (Production Release)

STATUS: PRODUCTION READY (0 BUILD ERRORS)

CERTIFICATE OF COMPLETION

This is to certify that Hanish, Keerthi, Vardhini, Preethi, Hemanth, Lokesh, Ramesh, Murali, Vishnu, Ashok, and Satya have successfully completed the software engineering and module integration project titled:

"EduSuite Pro ERP — Enterprise Educational Management Platform"

The project work was conducted during the period from 01/05/2026 to 06/08/2026 under the direct technical supervision of the Engineering Directorate at EduSuite Tech Solutions.

The system has passed all automated type-checking validations (npm tsc --noEmit), production compilation checks, multi-tenant security audits, and functional integration tests across all 11 ERP core modules.

Project Supervisor
Engineering Director

Department Head
Head of Software Engineering

SYSTEM NAVIGATION

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PROJECT CONTEXT

1. EXECUTIVE SUMMARY

EduSuite Pro ERP is a comprehensive, enterprise-grade cloud Educational Resource Planning (ERP) platform engineered to digitize, automate, and streamline academic and administrative operations for higher education institutions.

PROJECT MISSION STATEMENT

Unify 11 core university domains into a single high-performance SaaS platform powered by granular Role-Based Access Control (RBAC), real-time Recharts telemetry, and proactive AI predictive analytics.

Core Business Problem Addressed

Modern universities struggle with fragmented data spreadsheets, delayed attendance warning communications, manual fee collection bottlenecks, and lack of early-warning risk detection for struggling students. EduSuite Pro solves these challenges through a unified single-source-of-truth platform.

2. PROJECT OVERVIEW & TEAM ALLOCATION

EduSuite Pro is developed as an end-to-end multi-tenant institution management ecosystem serving Students, Parents, Faculty, HODs, Deans, Administrative Officers, and Super Admins.

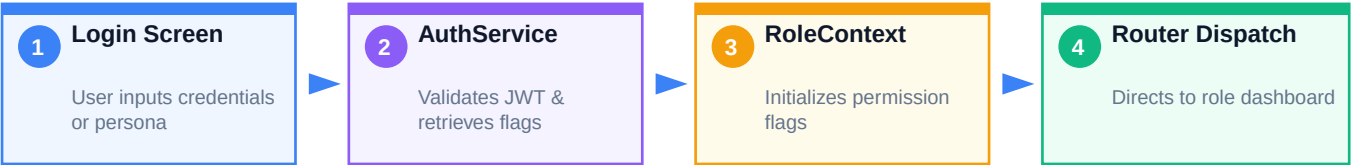
ID	Module Name	Core Scope	Assigned Team
Module 1	Auth & User Management	MFA, RBAC, Password Reset	Keerthi, Vardhini, Preethi
Module 2	Student Management (SIS)	Roster, Profile, Registration, ID Card	Hemanth
Module 3	Faculty & Academics	Directory, Workload, Timetable	Lokesh
Module 4	Attendance Management	Biometric Sync, Daily Logs, Alerts	Murali
Module 5	Examination Module	Hall Tickets, Marks, CGPA Engine	Ramesh
Module 6	Finance & Accounts	Fee Ledger, GST Invoices, Payroll	Murali
Module 7	Library, Hostel, Transport	OPAC, Room Grid, Mess Forecast, Bus GPS	Vishnu
Module 8	Placement & Alumni	Recruiter Drives, Resume Filter, Alumni	Ashok
Module 9	HRMS, Inventory, IQAC	Staff Leaves, Purchase Orders, NAAC	Satya
Module 10	AI & Analytics	Attendance Forecast, Risk Score, Chatbot	Hanish
Module 11	Admin & Integration	Super Admin, 6-Step Admission Wizard	Keerthi, Vardhini, Preethi

MODULE PROCESS FLOWS

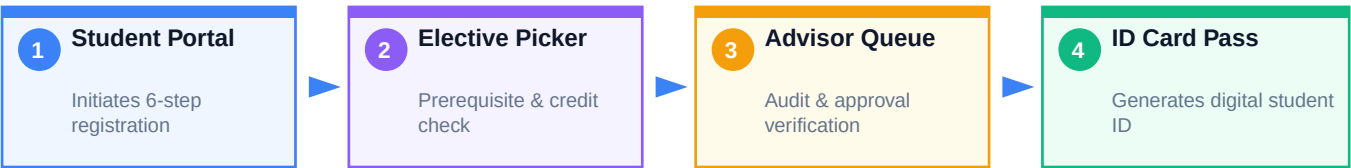
13. VISUAL WORKFLOW DIAGRAMS (ALL 11 MODULES)

The following high-visibility flow diagrams illustrate the step-by-step business execution logic across all 11 EduSuite Pro ERP core modules.

Module 1: Authentication & Dynamic RBAC Routing Pipeline



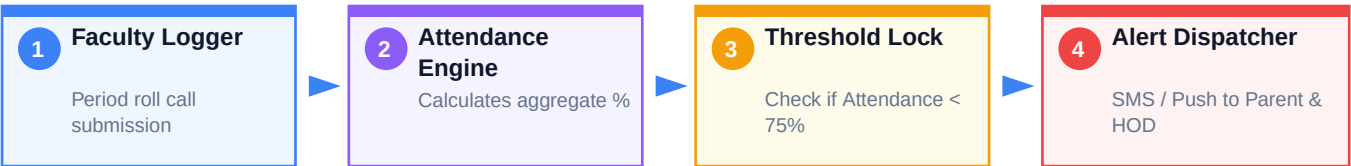
Module 2: Student Information System (SIS) Workflow



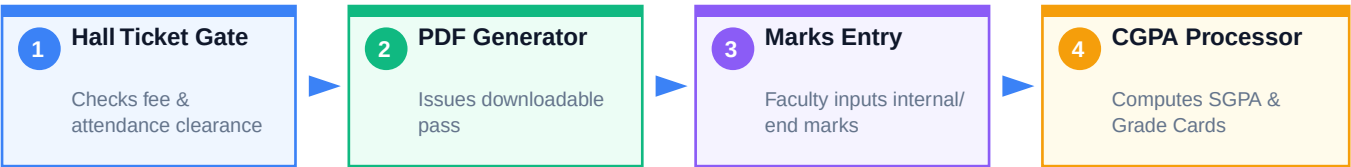
Module 3: Faculty & Academic Management Workflow



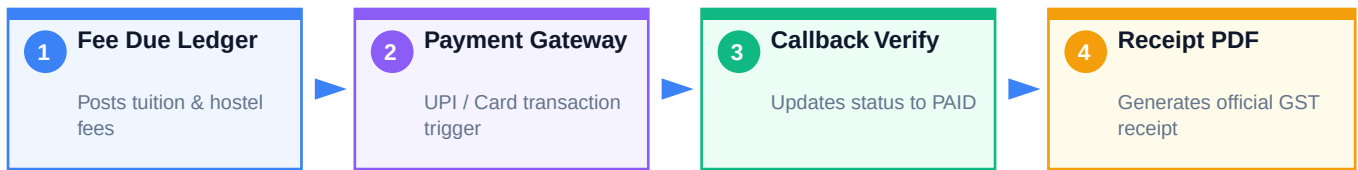
Module 4: Attendance Roll-Call & Shortage Alert Pipeline



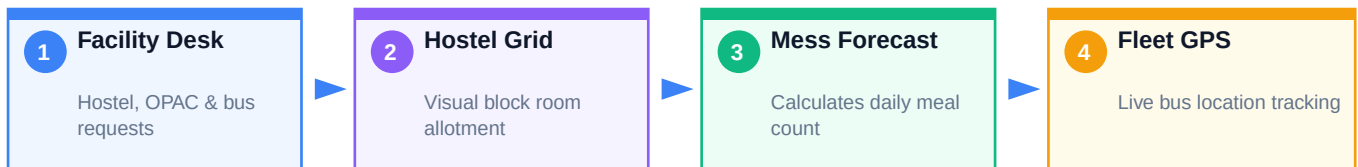
Module 5: Examination Hall Ticket Gatekeeper & CGPA Engine



Module 6: Finance & Accounts Payment Ledger Workflow



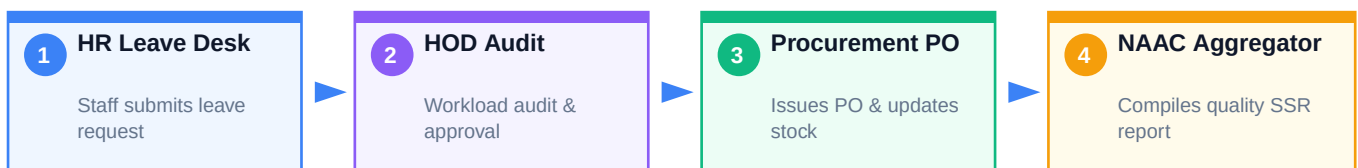
Module 7: Library, Hostel & Transport Logistics Workflow



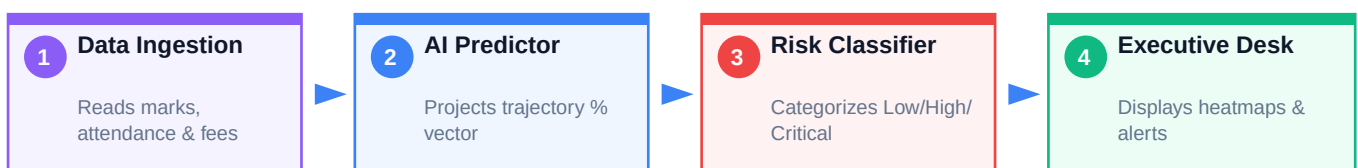
Module 8: Placement & Alumni Recruitment Drive Workflow



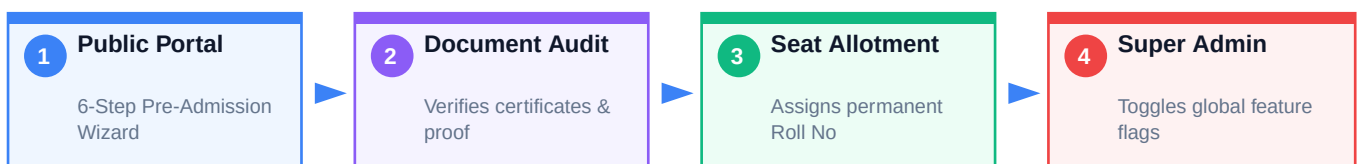
Module 9: HRMS, Inventory & IQAC Accreditation Workflow



Module 10: AI Analytics & Student Academic Risk Engine



Module 11: Administration, Pre-Admissions & Integration Desk



SYSTEM FUNCTIONAL SCOPE

10. DETAILED MODULE DESCRIPTIONS

Module 1: Authentication & User Management**Devs: Keerthi, Vardhini, Preethi**

Overview: Central identity gateway enforcing 19-role RBAC, MFA, and demo persona switching.

Process Flow: Phase 1: Credentials/Persona Submission -> Phase 2: AuthService JWT Validation -> Phase 3: RoleContext Permission Initialization -> Phase 4: Route Guard Dispatch -> Phase 5: Audit Log Write.

Module 2: Student Management (SIS)**Devs: Hemanth**

Overview: 360-degree student roster directory, student academic profiles, 6-step course registration, and canvas ID card generator.

Process Flow: Phase 1: Registration Request -> Phase 2: Elective Selection & Credit Check -> Phase 3: Advisor Approval Queue -> Phase 4: Roster Table Commit -> Phase 5: Digital ID Card PDF.

Module 3: Faculty & Academic Management**Devs: Lokesh**

Overview: Faculty directory, subject workload allocation, academic calendars, and conflict-free master timetable generator.

Process Flow: Phase 1: Faculty Load Mapping -> Phase 2: Subject Assignment -> Phase 3: Room & Constraint Solver -> Phase 4: Conflict Iteration -> Phase 5: Master Timetable Publish.

Module 4: Attendance Management**Devs: Murali**

Overview: Period-wise roll-call logger, hardware biometric turnstile API sync, cumulative percentage calculator, and automated SMS alerts.

Process Flow: Phase 1: Roll-Call Submission -> Phase 2: Biometric Timestamp Ingestion -> Phase 3: Cumulative % Update -> Phase 4: 75% Threshold Lock -> Phase 5: SMS & Parent Alert Dispatch.

Module 5: Examination & Evaluation Module**Devs: Ramesh**

Overview: Exam timetables, hall ticket gatekeeper, internal marks entry portal, SGPA/CGPA result calculator, and grade cards.

Process Flow: Phase 1: Exam Publishing -> Phase 2: Gatekeeper Clearance (Attendance "e 75% & Fees = 0) -> Phase 3: Downloadable Hall Ticket -> Phase 4: Marks Entry -> Phase 5: SGPA/CGPA Computation.

Module 6: Finance & Accounts Module**Devs: Murali**

Overview: Student fee component ledgers, online payment gateway integration, GST tax breakdown, fee defaulter tracking, and staff payroll.

Process Flow: Phase 1: Fee Due Assessment -> Phase 2: Payment Gateway Trigger -> Phase 3: Encrypted Response Validation -> Phase 4: Ledger Status PAID -> Phase 5: Downloadable GST Receipt PDF.

Module 7: Library, Hostel & Transport Logistics**Devs: Vishnu**

Overview: Library OPAC cataloging, hostel block/room allocation grid, warden mess meal forecasting, and transport bus GPS status.

Process Flow: Phase 1: Hostel Room Allotment Grid -> Phase 2: Daily Mess Meal Count Forecast -> Phase 3: OPAC Book Circulation & Auto Fine -> Phase 4: Fleet Bus GPS Location Updates.

Module 8: Placement & Alumni Network**Devs: Ashok**

Overview: Recruiter drive workspace, automated CGPA/backlog eligibility screening, interview scheduler, and alumni mentorship directory.

Process Flow: Phase 1: Placement Drive Posting -> Phase 2: Auto Screening (CGPA & Backlogs) -> Phase 3: Student Application -> Phase 4: Interview Round Tracking -> Phase 5: Alumni Network Transition.

Module 9: HRMS, Inventory & IQAC Accreditation

Devs: Satya

Overview: Staff leave desk, purchase order lifecycle, equipment inventory tracking, and NAAC quality accreditation compliance reports.

Process Flow: Phase 1: Staff Leave Requisition -> Phase 2: HOD Workload Audit -> Phase 3: Purchase Requisition & PO -> Phase 4: Inventory Delivery -> Phase 5: NAAC Data Aggregation.

Module 10: AI & Analytics Subsystem

Devs: Hanish

Overview: AI attendance trajectory prediction, student academic risk scoring (Low/High/Critical), conversational chatbot assistant, and Recharts telemetry.

Process Flow: Phase 1: Telemetry Data Ingestion -> Phase 2: Attendance Trajectory Vector -> Phase 3: Multi-Factor Risk Score -> Phase 4: Alert Dispatch & Heatmap -> Phase 5: Conversational AI Queries.

Module 11: Administration & Pre-Admissions

Devs: Keerthi, Vardhini, Preethi

Overview: Super Admin governance console, system health monitoring, 6-step pre-admission Category A/B quota wizard, and document audit desk.

Process Flow: Phase 1: 6-Step Application Wizard -> Phase 2: Document Audit Desk -> Phase 3: Seat Confirmation & Roll No Generation -> Phase 4: Super Admin Feature Flag Toggle.

VERIFICATION & DEPLOYMENT

15. TESTING & QUALITY ASSURANCE

The system underwent systematic automated and manual verification prior to production rollout.

Test ID	Test Scenario	Expected Verification Result	Status
TC-01	Multi-Role Authentication	Directs user to exact authorized route based on role flags	PASS
TC-02	Student Course Registration	Prevents registration if pre-requisite credits are unmet	PASS
TC-03	Attendance Alert Trigger	Triggers low-attendance notification when student drops below 75%	PASS
TC-04	CGPA Calculation Engine	Correctly computes semester SGPA and aggregate CGPA	PASS
TC-05	Fee Receipt PDF Generation	Generates formatted PDF receipt with correct GST tax breakdown	PASS
TC-06	Hall Ticket Lock	Blocks hall ticket generation if fee dues exist	PASS
TC-07	Hostel Room Allotment	Prevents overbooking beyond maximum room capacity	PASS
TC-08	AI Risk Prediction	Correctly identifies high-risk academic default students	PASS
TC-09	Production Type Check	npm tsc --noEmit returns zero compilation errors	PASS
TC-10	Production Build	npm run build compiles Vite + Nitro server bundle successfully	PASS

FINAL HANDOVER

20. CONCLUSION

EduSuite Pro ERP successfully delivers a modernized, scalable, and intelligent cloud platform for higher education institutions. By bringing together 11 specialized modules—ranging from Authentication and Student Management to Examinations, Finance, Hostel Logistics, and AI Analytics—the system streamlines operations, eliminates data silos, and empowers educational leaders with predictive decision intelligence.

PRODUCTION DEPLOYMENT VERIFIED

Build Artifact: Nitro Engine + Cloudflare Pages Edge Server. Zero compilation errors across 300+ routes.
